

Ayush Patel

📍 Mumbai | ✉ ayushptl1810@gmail.com | 📞 +91 7208494565 | 🔗 LinkedIn | 🌐 GitHub | 🌐 Portfolio

Education

SVKM's Dwarkadas Jivanlal Sanghvi College of Engineering	August 2023
BTech in Computer Science and Engineering (Data Science)	CGPA: 9.27
IIT Madras	June 2023
BS in Data Science	CGPA: 8.1

Skills

Programming Languages: Python, PostgreSQL, MySQL, Shell Scripting
Web Technologies & Tools: Javascript, ReactJs, NextJs, NodeJs, ExpressJs, Flask, FastAPI, MongoDB, Websockets
ML & Data Science: PyTorch, TensorFlow, Scikit-Learn, Pandas, Numpy, Matplotlib, Seaborn, OpenCV, Wandb
AI/Agentic: LangChain, LangGraph, CrewAI, HF Transformers, Vector Dbs, Graph Dbs, ONNX, Ollama
DevOps, Cloud & Automation: Docker, Git, Ngrok, Vercel, Render, HuggingFace Spaces, Playwright, Selenium

Projects

- 1. Agentic Data Intelligence System** | CrewAI, LangChain, FastAPI, Pinecone, ONNX *August 2025*
 - **Architected an autonomous multi-agent AI system** that decomposes natural language queries into multi-step plans, dynamically selects tools (SQL, PDF, web, vector search), and executes chained analytical workflows.
 - **Built scalable semantic retrieval infrastructure** using Pinecone vector embeddings and ONNX-optimized models for high-performance chunking, similarity search, and retrieval-augmented analysis.
 - **Exposed the agent as a production-ready FastAPI service**, supporting structured responses, automated data visualizations, and Pandas-driven transformation pipelines for real-time analytical insight generation.
- 2. Samvidha – Constitutional Intelligence Agent** | ReactJs, LangGraph, Neo4j, Qdrant, FastAPI *Mar 2026*
 - **Built a GraphRAG multi-agent system** using LangGraph with parallel Neo4j graph traversal and Qdrant vector search, enabling multi-hop Cypher queries to trace cross-article constitutional relationships and amendments.
 - **Engineered a self-validating hybrid retrieval pipeline** with a strict "Hierarchy of Truth". Graph relationships override semantic chunks, and a critique agent recursively refines responses until citation accuracy is verified.
 - **Achieved sub-second agentic latency** by orchestrating parallel Neo4j/Qdrant retrieval with Groq's fast LLM inference and real-time SSE streaming.
- 3. Project Aegis – AI-Powered Fact-Checking Platform** | FastAPI, ReactJs, Websockets, Redis *Nov 2025*
 - **Built an end-to-end AI verification system** supporting text, image, video, and audio fact-checking using hybrid pipelines integrating Google Custom Search, SerpApi reverse image search, and an LLM for evidence synthesis.
 - **Developed real-time misinformation monitoring** via WebSockets and MongoDB change streams, enabling live rumour feed updates with intelligent filtering and TF-IDF-based similarity detection for deduplication.
 - **Created Chrome browser extension** that monitors web search queries in real-time, performs TF-IDF similarity matching against verified rumours database, displays contextual alerts with verdict badges and similarity scores and integrates context menu functionality for on-demand text verification.
- 4. Asynchronous Robust Federated Learning** | ReactJs, FastAPI, PyTorch *Feb 2026*
 - **Built a distributed training framework** addressing stragglers and Byzantine failures using staleness-aware weighting, robust aggregation (Trimmed Mean, Median), and secure aggregation via zero-sum masking.
 - **Implemented a two-layer Byzantine defense** using L2-norm anomaly filtering and SABD trust scoring – fusing behavioral reputation via robust Z-scores and staleness penalties to dynamically weigh client contributions.

Leadership & Engagement

- Vice Captain of **DJS Phoenix**, the official collegiate drone team.
- Selected for the **JPMorgan Chase External Engagement Program** for career-oriented skill development.
- **Regional Finalist at Aavishkar '25** and **3rd at DJS Sanshodhan** for EEG/EMG-based hunger state prediction.
- Awarded **2nd Runner-Up at DevHacks '26** among 200+ participating teams.
- **Winner at IndiaNext Hackathon, BuildStorm Track** at KES Shroff College.
- **1st Runner-Up at Codefolio '26**, a Portfolio Creation Competition at **SPIT**.